# Capstone Checkpoint 5.1 - Agentic TOC Retriever and Tables As_IS

Oracle, vector, and hybrid experiment comparison. Generated 2026-09-22 09:35 EDT.

| 11 | 3 | 209 | 19 | 0 |
|---|---|---|---|---|
| Included runs | Configurations | Question-runs | Questions | Legacy runs excluded |

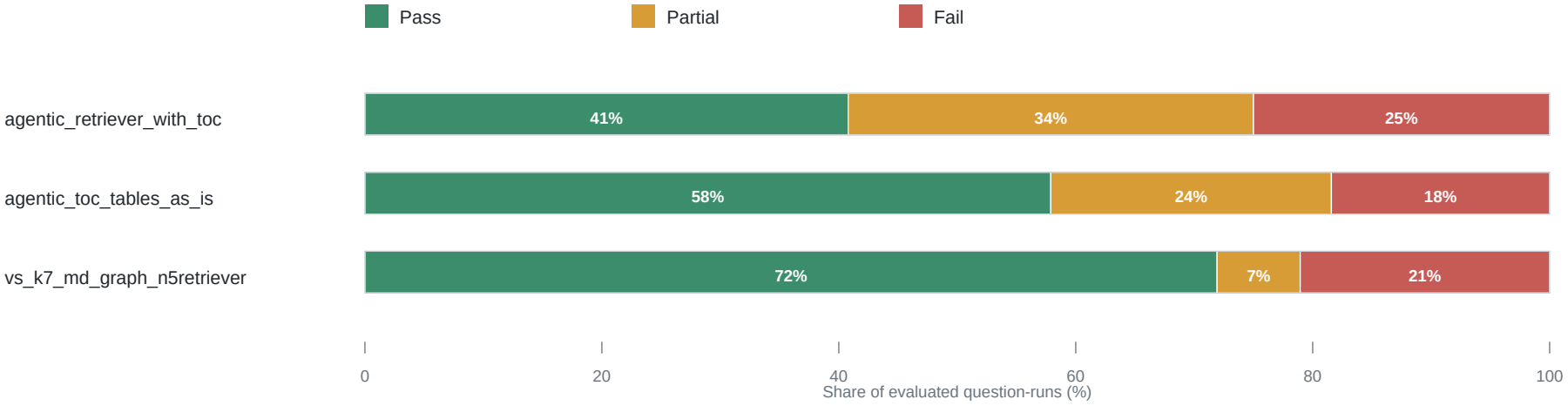
## Summary

The highest observed mean quality score was 0.754 for vs_k7_md_graph_n5retriever across 3 run(s). Quality assigns 1 to pass, 0.5 to partial, and 0 to fail.

The highest observed mean designated-source recall was 97.4% for agentic_retriever_with_toc. Article recall is recomputed from the required and retrieved article names in every CSV.

The highest task-family attention score was 73.3 for Exhaustive list or time filter under vs_k7_md_graph_n5retriever. Attention is 100 x (1 - grouped quality), so higher values identify larger observed improvement opportunities.

### Answer outcomes

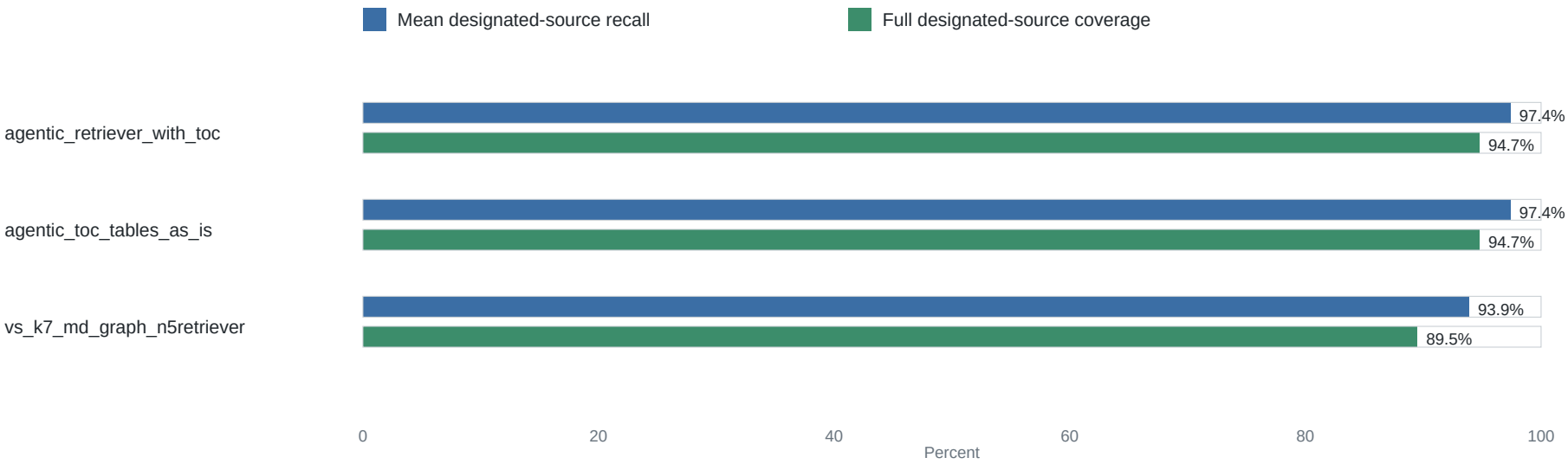



| Configuration | Runs | Cases | Pass | Partial | Fail | Quality | Run SD | Article recall | Full source coverage | Mean context |
|---|---|---|---|---|---|---|---|---|---|---|
| agentic_retriever_with_toc | 4 | 76 | 40.8% | 34.2% | 25.0% | 0.579 | 0.032 | 97.4% | 94.7% | 13.3k |
| agentic_toc_tables_as_is | 4 | 76 | 57.9% | 23.7% | 18.4% | 0.697 | 0.023 | 97.4% | 94.7% | 28.3k |
| vs_k7_md_graph_n5retriever | 3 | 57 | 71.9% | 7.0% | 21.1% | 0.754 | 0.045 | 93.9% | 89.5% | 260.7k |

*Percentages aggregate all question-runs within a configuration. Run SD is the standard deviation of mean quality across repeated runs.*

# Retrieval diagnostics

## Designated-source coverage




*The analyzer recomputes recall after removing only final .txt and .html suffixes. This corrects historical .html.txt versus .html mismatches. Designated-source recall is diagnostic: alternative retrieved articles may still contain sufficient evidence.*

## Outcome and source-coverage combinations

| Configuration | Pass, full | Pass, incomplete | Non-pass, full | Non-pass, missing |
|---|---|---|---|---|
| agentic_retriever_with_toc | 31 | 0 | 41 | 4 |
| agentic_toc_tables_as_is | 40 | 4 | 32 | 0 |
| vs_k7_md_graph_n5retriever | 38 | 3 | 13 | 3 |

A non-pass result with full designated-source recall points toward answer synthesis, context noise, table interpretation, or judge behavior. A non-pass result with a missing designated source is consistent with a retrieval recall problem, but manual review is still required.

# Per-question comparison

Cells show the mean quality score for each question and configuration across repeated runs.

| | agentic_retriever_with_toc | agentic_toc_tables_as_is | vs_k7_md_graph_n5retriever |
|---|---|---|---|
| GT01 | **1.00** | **1.00** | **1.00** |
| GT02 | **0.62** | **0.75** | **1.00** |
| GT03 | **0.00** | **0.12** | **0.00** |
| GT04 | **1.00** | **1.00** | **1.00** |
| GT05 | **1.00** | **1.00** | **1.00** |
| GT06 | **0.25** | **0.12** | **1.00** |
| GT07 | **0.25** | **0.50** | **0.33** |
| GT08 | **0.00** | **0.62** | **0.67** |
| GT09 | **0.38** | **0.75** | **1.00** |
| GT10 | **0.50** | **0.50** | **0.00** |
| GT11 | **0.00** | **0.12** | **0.00** |
| GT12 | **0.62** | **1.00** | **1.00** |
| GT13 | **0.88** | **1.00** | **1.00** |
| GT14 | **1.00** | **1.00** | **1.00** |
| GT15 | **1.00** | **1.00** | **1.00** |
| GT16 | **0.38** | **0.38** | **0.50** |
| GT17 | **0.62** | **0.50** | **1.00** |
| GT18 | **1.00** | **0.88** | **1.00** |
| GT19 | **0.50** | **1.00** | **0.83** |

Fail (0) Partial (0.5) Pass (1)

| Question | agentic_retriever_with_toc | agentic_toc_tables_as_is | vs_k7_md_graph_n5retriever |
|---|---|---|---|
| GT01 | 1.000 | 1.000 | 1.000 |
| GT02 | 0.625 | 0.750 | 1.000 |
| GT03 | 0.000 | 0.125 | 0.000 |
| GT04 | 1.000 | 1.000 | 1.000 |
| GT05 | 1.000 | 1.000 | 1.000 |
| GT06 | 0.250 | 0.125 | 1.000 |
| GT07 | 0.250 | 0.500 | 0.333 |
| GT08 | 0.000 | 0.625 | 0.667 |
| GT09 | 0.375 | 0.750 | 1.000 |
| GT10 | 0.500 | 0.500 | 0.000 |
| GT11 | 0.000 | 0.125 | 0.000 |
| GT12 | 0.625 | 1.000 | 1.000 |
| GT13 | 0.875 | 1.000 | 1.000 |
| GT14 | 1.000 | 1.000 | 1.000 |
| GT15 | 1.000 | 1.000 | 1.000 |
| GT16 | 0.375 | 0.375 | 0.500 |
| GT17 | 0.625 | 0.500 | 1.000 |
| GT18 | 1.000 | 0.875 | 1.000 |
| GT19 | 0.500 | 1.000 | 0.833 |

## Per-question article recall

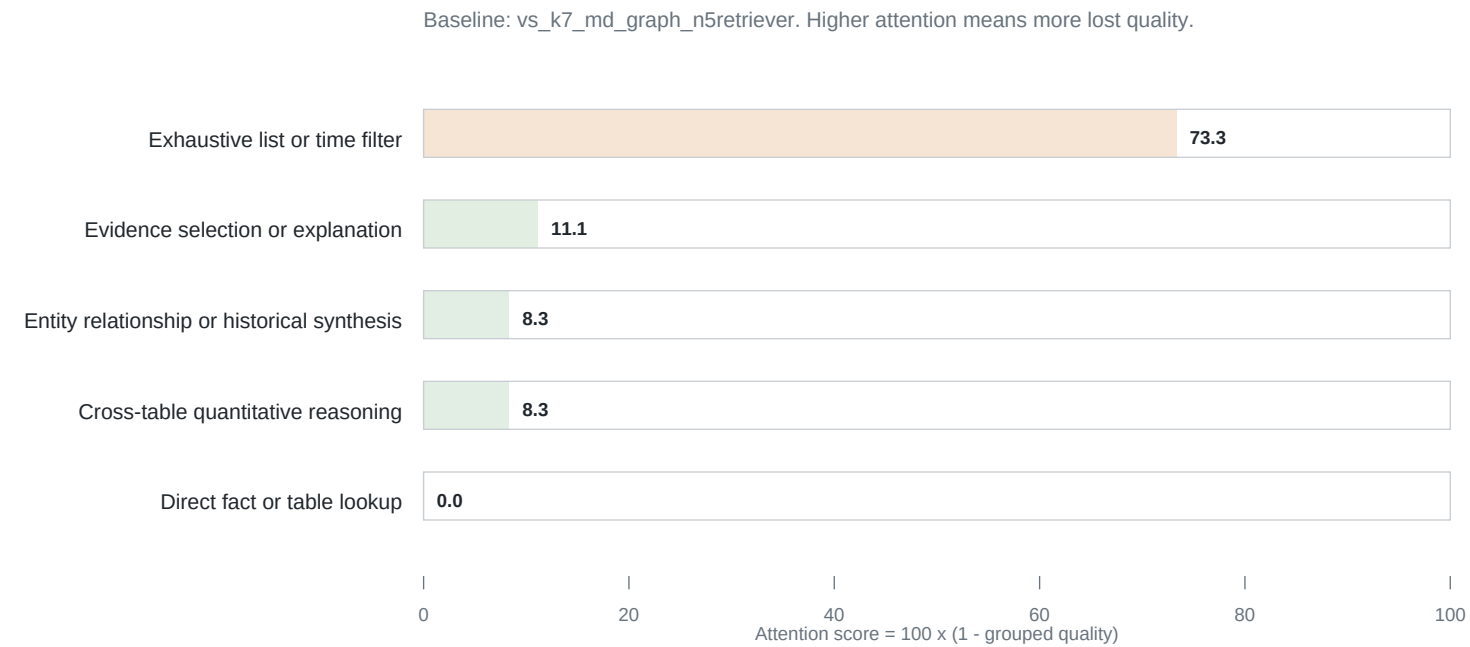
Fraction of each question's designated source articles that appeared in the retrieved set (retriever configurations only). 1.00 = every gold article retrieved; lower values mark a retrieval miss — usually a multi-source question.

| Question | agentic_retriever_with_toc | agentic_toc_tables_as_is | vs_k7_md_graph_n5retriever |
|---|---|---|---|
| GT01 | 100.0% | 100.0% | 100.0% |

| Question | agentic_retriever_with_toc | agentic_toc_tables_as_is | vs_k7_md_graph_n5retriever |
|---|---|---|---|
| GT02 | 100.0% | 100.0% | 100.0% |
| GT03 | 100.0% | 100.0% | 33.3% |
| GT04 | 100.0% | 100.0% | 100.0% |
| GT05 | 100.0% | 100.0% | 100.0% |
| GT06 | 100.0% | 100.0% | 100.0% |
| GT07 | 100.0% | 100.0% | 100.0% |
| GT08 | 100.0% | 100.0% | 100.0% |
| GT09 | 100.0% | 100.0% | 50.0% |
| GT10 | 100.0% | 100.0% | 100.0% |
| GT11 | 100.0% | 100.0% | 100.0% |
| GT12 | 100.0% | 100.0% | 100.0% |
| GT13 | 100.0% | 100.0% | 100.0% |
| GT14 | 100.0% | 100.0% | 100.0% |
| GT15 | 100.0% | 100.0% | 100.0% |
| GT16 | 100.0% | 100.0% | 100.0% |
| GT17 | 100.0% | 100.0% | 100.0% |
| GT18 | 100.0% | 100.0% | 100.0% |
| GT19 | 50.0% | 50.0% | 100.0% |

# Performance by task type

The 18 original category labels are consolidated into five task families so each group contains multiple questions. The attention score ranks observed improvement opportunities for the strongest retrieval configuration; it is not a statistical significance test.




| Task family | Questions | Baseline quality | Attention | Fail | Oracle quality | Oracle gap | Article recall |
|---|---|---|---|---|---|---|---|
| Exhaustive list or time filter | 5 | 0.267 | 73.3 | 66.7% | n/a | n/a | 86.7% |
| Evidence selection or explanation | 3 | 0.889 | 11.1 | 11.1% | n/a | n/a | 100.0% |
| Entity relationship or historical synthesis | 6 | 0.917 | 8.3 | 5.6% | n/a | n/a | 100.0% |
| Cross-table quantitative reasoning | 2 | 0.917 | 8.3 | 0.0% | n/a | n/a | 75.0% |
| Direct fact or table lookup | 3 | 1.000 | 0.0 | 0.0% | n/a | n/a | 100.0% |

*Baseline quality and failure rate use vs_k7_md_graph_n5retriever. Oracle gap equals oracle quality minus baseline quality: a large positive gap is consistent with a retrieval or context-selection opportunity, while low oracle quality indicates that the difficulty persists even when the designated articles are supplied.*

# Performance by source scope

Source scope groups questions by the number of designated articles. It tests whether the number of sources, rather than the reasoning operation, explains performance.

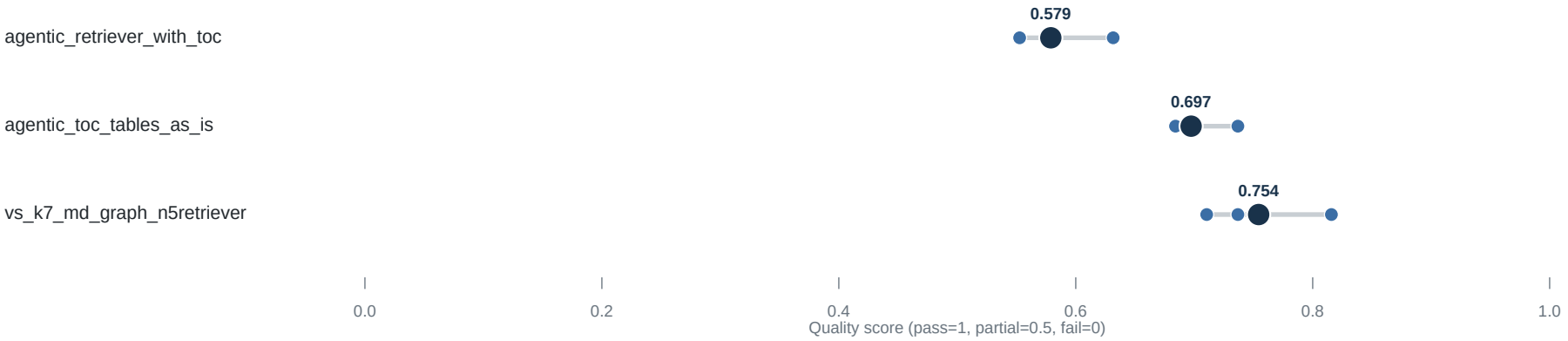
| | agentic_retriever_with_toc | agentic_toc_tables_as_is | vs_k7_md_graph_n5retriever |
|---|---|---|---|
| 1 article | **0.591** | **0.682** | **0.727** |
| 2 articles | **0.575** | **0.725** | **0.867** |
| 3+ articles | **0.542** | **0.708** | **0.667** |

| Source scope | Configuration | Questions | Question-runs | Pass | Partial | Fail | Quality | Article recall |
|---|---|---|---|---|---|---|---|---|
| 1 article | agentic_retriever_with_toc | 11 | 44 | 47.7% | 22.7% | 29.5% | 0.591 | 100.0% |
| 2 articles | agentic_retriever_with_toc | 5 | 20 | 25.0% | 65.0% | 10.0% | 0.575 | 90.0% |
| 3+ articles | agentic_retriever_with_toc | 3 | 12 | 41.7% | 25.0% | 33.3% | 0.542 | 100.0% |
| 1 article | agentic_toc_tables_as_is | 11 | 44 | 59.1% | 18.2% | 22.7% | 0.682 | 100.0% |
| 2 articles | agentic_toc_tables_as_is | 5 | 20 | 50.0% | 45.0% | 5.0% | 0.725 | 90.0% |
| 3+ articles | agentic_toc_tables_as_is | 3 | 12 | 66.7% | 8.3% | 25.0% | 0.708 | 100.0% |
| 1 article | vs_k7_md_graph_n5retriever | 11 | 33 | 69.7% | 6.1% | 24.2% | 0.727 | 100.0% |
| 2 articles | vs_k7_md_graph_n5retriever | 5 | 15 | 80.0% | 13.3% | 6.7% | 0.867 | 90.0% |
| 3+ articles | vs_k7_md_graph_n5retriever | 3 | 9 | 66.7% | 0.0% | 33.3% | 0.667 | 77.8% |

*Question-runs are repeated measurements of the listed unique questions. Source-scope groups remain small and mix different task families, so this view should be used as a diagnostic rather than a general performance estimate.*

# Run-to-run variability

Each dot is one run; the larger dot is the configuration mean.




| Run | Configuration | Cases | Pass | Partial | Fail | Quality | Article recall | Full source coverage |
|---|---|---|---|---|---|---|---|---|
| eloquent_bezos | agentic_retriever_with_toc | 19 | 9 | 3 | 7 | 0.553 | 97.4% | 94.7% |
| trusting_turing | agentic_retriever_with_toc | 19 | 7 | 7 | 5 | 0.553 | 97.4% | 94.7% |
| zen_cerf | agentic_retriever_with_toc | 19 | 7 | 8 | 4 | 0.579 | 97.4% | 94.7% |
| naughty_carmack | agentic_retriever_with_toc | 19 | 8 | 8 | 3 | 0.632 | 97.4% | 94.7% |
| sweet_hopper | agentic_toc_tables_as_is | 19 | 10 | 6 | 3 | 0.684 | 97.4% | 94.7% |
| eloquent_lovelace | agentic_toc_tables_as_is | 19 | 11 | 4 | 4 | 0.684 | 97.4% | 94.7% |
| brave_bezos | agentic_toc_tables_as_is | 19 | 11 | 4 | 4 | 0.684 | 97.4% | 94.7% |
| hardcore_wing | agentic_toc_tables_as_is | 19 | 12 | 4 | 3 | 0.737 | 97.4% | 94.7% |
| bold_wozniak | vs_k7_md_graph_n5retriever | 19 | 13 | 2 | 4 | 0.737 | 93.9% | 89.5% |
| awesome_valiant | vs_k7_md_graph_n5retriever | 19 | 13 | 1 | 5 | 0.711 | 93.9% | 89.5% |
| reverent_wilkes | vs_k7_md_graph_n5retriever | 19 | 15 | 1 | 3 | 0.816 | 93.9% | 89.5% |

# Methodology and limitations

## Scoring

Pass is assigned 1, partial is assigned 0.5, and fail is assigned 0. The quality score is the arithmetic mean across the applicable question-runs. Pass, partial, and fail percentages are also reported separately so the weighted score does not hide the underlying verdicts.

## Task-family attention

The attention score is 100 multiplied by one minus grouped quality. A score of 66 means that the group lost 66% of the available quality points under the selected retrieval baseline. It is a prioritization aid, not an estimate of population-wide failure risk. Task-family and source-scope results must be read with their unique-question counts because the evaluation set contains only 19 questions.

## Configuration grouping

Runs are grouped by a CSV experiment_name or config_name when one is present. If no explicit name is present, grouping falls back to evaluation mode, context source, retriever kind and label, Top K, hybrid weights, answer model, and judge model. RAGAS-generated filenames identify individual runs but do not define configurations.

## Framework compatibility

Legacy oracle runs stored under pass_fail_only directories are excluded by default because they could not produce partial verdicts. Use --include-binary-oracle only for a labeled historical appendix, not for a direct three-level score comparison.

## Interpretation limits

An LLM produced the answers and another LLM applied the rubric, so repeated runs can vary. Article recall measures coverage of designated ground-truth sources, not every possible supporting source. Full article recall does not prove that the model used the evidence correctly, and missing a designated source does not prove the context was insufficient when another article contains equivalent evidence. Differences between configurations should be interpreted together with run counts and per-question behavior.

## Included experiment files

| Run file |
|---|
| awesome_valiant.csv |
| bold_wozniak.csv |
| brave_bezos.csv |
| eloquent_bezos.csv |
| eloquent_lovelace.csv |
| hardcore_wing.csv |
| naughty_carmack.csv |
| reverent_wilkes.csv |
| sweet_hopper.csv |
| trusting_turing.csv |
| zen_cerf.csv |